Literature search report – run 20260828T100120

scitech-librarian 3.1

August 28, 2026

Generated by scitech-librarian 3.1 (report level: simple). Every number below is reproducible from the archived run directory `20260828T100120`.

Item	Value
Run started	2026-08-28 10:01:20
Duration	128 s
Query file	queries.example.json
Blocks	PSC, MLP, NOV, QC
Backends	openalex, arxiv, inspire, semanticscholar, ads, scopus
Mode	full fetch, up to 300 records per block and backend
Non-curated venue filter	on
Open-access lookup	not run
Interrupted	no

Search strategy

Each block is one structural query – a conjunction of synonym groups, (a OR b) AND (c OR d) – rendered into every backend's native grammar. The strings below are exactly what was sent (PRISMA-S item 8).

Block PSC: Perovskite solar cell degradation under humidity

Purpose: grounding block – expect thousands of hits; use it to sanity-check that every backend is reachable and the counts are plausible

```
(perovskite solar cell OR halide perovskite photovoltaic) AND (degradation OR stability OR encapsulation)
 AND (humidity OR moisture OR water ingress)
```

Backend	Query string sent
openalex	("perovskite solar cell" OR "halide perovskite photovoltaic") AND (degradation OR stability OR encapsulation) AND (humidity OR moisture OR "water ingress")
arxiv	(all:"perovskite solar cell" OR all:"halide perovskite photovoltaic") AND (all:"degradation" OR all:"stability" OR all:"encapsulation")
inspire	("perovskite solar cell" or "halide perovskite photovoltaic") and ("degradation" or "stability" or "encapsulation") and ("humidity" or "moisture" or "water ingress")
semanticscholar	("perovskite solar cell" | "halide perovskite photovoltaic") + ("degradation" | "stability" | "encapsulation") + ("humidity" | "moisture" | "water ingress")
ads	(abs:"perovskite solar cell" OR abs:"halide perovskite photovoltaic") AND (abs:"degradation" OR abs:"stability" OR abs:"encapsulation") AND (abs:"humidity" OR abs:"moisture" OR abs:"water ingress")
scopus	TITLE-ABS-KEY("perovskite solar cell" OR "halide perovskite photovoltaic") AND TITLE-ABS-KEY(degradation OR stability OR encapsulation) AND TITLE-ABS-KEY(humidity OR moisture OR "water ingress")

Block MLP: Machine-learned interatomic potentials for amorphous materials

Purpose: a narrower intersection – expect hundreds; good for testing record fetch and RIS export

```
(machine learning potential OR machine-learned interatomic potential OR neural network potential) AND
(amorphous OR glass OR disordered solid) AND (molecular dynamics OR structure prediction)
```

arXiv receives groups [0, 1] only (nested-boolean limitation).

Backend	Query string sent
openalex	("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND (amorphous OR glass OR "disordered solid") AND ("molecular dynamics" OR "structure prediction")
arxiv	(all:"machine learning potential" OR all:"machine-learned interatomic potential" OR all:"neural network potential") AND (all:"amorphous" OR all:"glass" OR all:"disordered solid")
inspire	("machine learning potential" or "machine-learned interatomic potential" or "neural network potential") and ("amorphous" or "glass" or "disordered solid") and ("molecular dynamics" or "structure prediction")
semanticscholar	("machine learning potential" | "machine-learned interatomic potential" | "neural network potential") + ("amorphous" | "glass" | "disordered solid") + ("molecular dynamics" | "structure prediction")
ads	(abs:"machine learning potential" OR abs:"machine-learned interatomic potential" OR abs:"neural network potential") AND (abs:"amorphous" OR abs:"glass" OR abs:"disordered solid") AND (abs:"molecular dynamics" OR abs:"structure prediction")
scopus	TITLE-ABS-KEY("machine learning potential" OR "machine-learned interatomic potential" OR "neural network potential") AND TITLE-ABS-KEY(amorphous OR glass OR "disordered solid") AND TITLE-ABS-KEY("molecular dynamics" OR "structure prediction")

Block NOV: EXAMPLE NOVELTY CHECK: origami metamaterials as topological acoustic pumps

Purpose: novelty-check pattern – a SMALL number is the good outcome; read every hit by hand before claiming a gap

```
(origami OR kirigami) AND (acoustic metamaterial OR phononic crystal) AND (topological pumping OR
Thouless pump OR edge state)
```

Backend	Query string sent
openalex	(origami OR kirigami) AND ("acoustic metamaterial" OR "phononic crystal") AND ("topological pumping" OR "Thouless pump" OR "edge state")
arxiv	(all:"origami" OR all:"kirigami") AND (all:"acoustic metamaterial" OR all:"phononic crystal")
inspire	("origami" or "kirigami") and ("acoustic metamaterial" or "phononic crystal") and ("topological pumping" or "Thouless pump" or "edge state")
semanticscholar	("origami" | "kirigami") + ("acoustic metamaterial" | "phononic crystal") + ("topological pumping" | "Thouless pump" | "edge state")
ads	(abs:"origami" OR abs:"kirigami") AND (abs:"acoustic metamaterial" OR abs:"phononic crystal") AND (abs:"topological pumping" OR abs:"Thouless pump" OR abs:"edge state")
scopus	TITLE-ABS-KEY(origami OR kirigami) AND TITLE-ABS-KEY("acoustic metamaterial" OR "phononic crystal") AND TITLE-ABS-KEY("topological pumping" OR "Thouless pump" OR "edge state")

Block QC: Quasicrystal photonics (arXiv-tuned example)

Purpose: demonstrates arxiv_groups: arXiv chokes on deeply nested booleans, so name the two groups it should get

```
(quasicrystal OR quasiperiodic lattice) AND (photonic OR waveguide OR optical lattice) AND (localization
OR critical states OR fractal spectrum)
```

arXiv receives groups [0, 2] only (nested-boolean limitation).

Backend	Query string sent
openalex	(quasicrystal OR "quasiperiodic lattice") AND (photonic OR waveguide OR "optical lattice") AND (localization OR "critical states" OR "fractal spectrum")
arxiv	(all:"quasicrystal" OR all:"quasiperiodic lattice") AND (all:"localization" OR all:"critical states" OR all:"fractal spectrum")
inspire	("quasicrystal" or "quasiperiodic lattice") and ("photonic" or "waveguide" or "optical lattice") and ("localization" or "critical states" or "fractal spectrum")
semanticscholar	("quasicrystal" | "quasiperiodic lattice") + ("photonic" | "waveguide" | "optical lattice") + ("localization" | "critical states" | "fractal spectrum")
ads	(abs:"quasicrystal" OR abs:"quasiperiodic lattice") AND (abs:"photonic" OR abs:"waveguide" OR abs:"optical lattice") AND (abs:"localization" OR abs:"critical states" OR abs:"fractal spectrum")
scopus	TITLE-ABS-KEY(quasicrystal OR "quasiperiodic lattice") AND TITLE-ABS-KEY(photonic OR waveguide OR "optical lattice") AND TITLE-ABS-KEY(localization OR "critical states" OR "fractal spectrum")

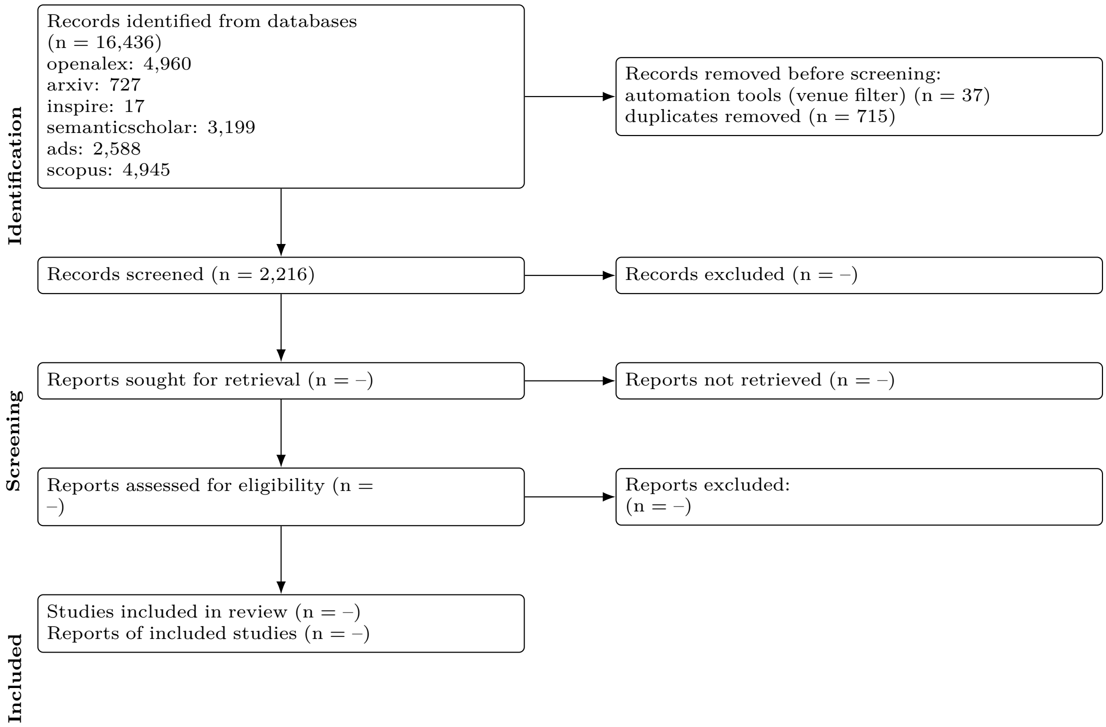
Results summary

Block	openalex	arxiv	inspire	semanticscholar	ads	scopus	Identified	Retrieved	Unique
PSC	4567	175	0	2968	2179	4731	14,620	1275	1163
MLP	239	138	0	109	203	149	838	802	444
NOV	0	0	0	0	0	0	0	0	0
QC	154	414	17	122	206	65	978	854	609

Block	openalex	arxiv	inspire	semanticscholar	ads	scopus	Identified	Retrieved	Unique
Total	4,960	727	17	3,199	2,588	4,945	16,436	2931	2216

Identified = database hit counts (not comparable across backends: proximity operators are dropped and stemming differs). Retrieved = records actually downloaded after the venue filter, capped by `--limit`. Unique = after DOI/title deduplication across all backends.

PRISMA 2020 flow



Stage	n
Records identified from databases	16,436
openalex	4,960
arxiv	727
inspire	17
semanticscholar	3,199
ads	2,588
scopus	4,945
Records retrieved (downloaded)	2,968
Removed before screening: automation (non-curated venues)	37
Removed before screening: duplicates	715
Records to screen (unique)	2,216
Records screened	2,216 (assumed = unique)
Records excluded at screening	–
Reports sought for retrieval	–
Reports not retrieved	–
Reports assessed for eligibility	–
Studies included	–
Reports of included studies	–

Automation stages are computed from the run; '–' marks manual stages not yet recorded in prisma.json. Note that 'identified' counts hits reported by each database while 'retrieved' is what was downloaded within `--limit`, so the two differ on large blocks.

PRISMA-S search-reporting checklist

Item	Requirement	This search
1	Database name	openalex, arxiv, inspire, semanticscholar, ads, scopus (queried through their documented public APIs)
2	Multi-database searching	6 databases, one structural query per block rendered into each native grammar; see Search strategy
3	Study registries	not applicable
4	Online resources and browsing	not applicable

Item	Requirement	This search
5	Citation searching	not performed by the tool (manual, if any)
6	Contacts	not applicable
7	Other methods	none
8	Full search strategies	reported verbatim per backend under Search strategy; archived in queries.json
9	Limits and restrictions	record download capped at 300 per block and backend, most-cited first; no date, language or document-type limits applied
10	Search filters	venue filter on: records from non-curated repositories (Zenodo, Figshare, SSRN...) removed
11	Prior work	none
12	Updates	2 earlier run(s) archived; counts tracked in counts_history.csv
13	Dates of searches	all databases searched on 2026-08-28 10:01:20
14	Peer review	none
15	Total records	16,436 identified; 2,968 retrieved; 2,216 unique
16	Deduplication	exact DOI match, else first 90 characters of the lower-cased title; 715 duplicates removed

Top 10 records per block

Deduplicated across backends, sorted by citation count. The complete set is in all_records.csv / .ris.

Block PSC (1163 unique)

Title	Authors	Year	Venue	Cited	DOI
Incorporation of rubidium cations into perovskite solar cells improves photovoltaic performance	Michael Saliba; Taisuke Matsui; Konrad Domanski et al.	2016	Science	3691	10.1126/science.aah5557
High-efficiency two-dimensional Ruddlesden–Popper perovskite solar cells	Hsinhan Tsai; Wanyi Nie; Jean-Christophe Blancon et al.	2016	Nature	3360	10.1038/nature18306
Efficient, stable and scalable perovskite solar cells using poly(3-hexylthiophene)	Eui Hyuk Jung; Nam Joong Jeon; Eun Young Park et al.	2019	Nature	2318	10.1038/s41586-019-1036-3
A Layered Hybrid Perovskite Solar-Cell Absorber with Enhanced Moisture Stability	Ian C. P. Smith; Eric T. Hoke; Diego Solís-Ibarra et al.	2014	Angewandte Chemie International Edition	2006	10.1002/anie.201406466
Review of recent progress in chemical stability of perovskite solar cells	Guangda Niu; Xudong Guo; Liduo Wang	2014	Journal of Materials Chemistry A	1942	10.1039/c4ta04994b
Understanding Degradation Mechanisms and Improving Stability of Perovskite Photovoltaics	Caleb C. Boyd; Rongrong Cheacharoen; Tomas Leijtens et al.	2018	Chemical Reviews	1767	10.1021/acs.chemrev.8b00336
Formamidinium and Cesium Hybridization for Photo- and Moisture-Stable Perovskite Solar Cell	Jin-Wook Lee; Deok-Hwan Kim; Hui-Seon Kim et al.	2015	Advanced Energy Materials	1661	10.1002/aenm.201501310
23.6%-efficient monolithic perovskite/silicon tandem solar cells with improved stability	Kevin A. Bush; Axel F. Palmstrom; Zhengshan J. Yu et al.	2017	Nature Energy	1533	10.1038/nenergy.2017.9
Investigation of CH 3 NH 3 PbI 3 Degradation Rates and Mechanisms in Controlled Humidity Environments Using in Situ Techniques	Jinli Yang; Braden D. Siempelkamp; Dianyi Liu et al.	2015	ACS Nano	1342	10.1021/nn506864k
Carbon Nanotube/Polymer Composites as a Highly Stable Hole Collection Layer in Perovskite Solar Cells	Severin N. Habisreutinger; Tomas Leijtens; Giles E. Eperon et al.	2014	Nano Letters	1198	10.1021/nl501982b

Block MLP (444 unique)

Title	Authors	Year	Venue	Cited	DOI
Atom-centered symmetry functions for constructing high-dimensional neural network potentials	Jörg Behler	2011	The Journal of Chemical Physics	1643	10.1063/1.3553717
Realistic Atomistic Structure of Amorphous Silicon from Machine-Learning-Driven Molecular Dynamics	Volker L. Deringer; Noam Bernstein; Albert P. Bartók et al.	2018	The Journal of Physical Chemistry Letters	269	10.1021/acs.jpclett.8b00902
Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide	G. Sivaraman; A. Krishnamoorthy; Matthias Baur et al.	2020	npj Computational Materials	195	10.1038/s41524-020-00367-7
Machine-learned interatomic potentials by active learning: amorphous and liquid hafnium dioxide	Ganesh Sivaraman; Anand Narayanan Krishnamoorthy; Matthias Baur et al.	2019	Cambridge University Engineering Department Publications Database	184	10.17863/cam.55408
Growth Mechanism and Origin of High sp3 Content in Tetrahedral Amorphous Carbon	Miguel A. Caro; Volker L. Deringer; Jari Koskinen et al.	2018	Physical Review Letters	178	10.1103/physrevlett.120.166101

Title	Authors	Year	Venue	Cited	DOI
Constructing first-principles phase diagrams of amorphous LixSi using machine-learning-assisted sampling with an evolutionary algorithm	Nongnuch Artrith; Alexander Urban; Gerbrand Ceder	2018	The Journal of Chemical Physics	155	10.1063/1.5017661
Study of Li atom diffusion in amorphous Li3PO4 with neural network potential	Wenwen Li; Yasunobu Ando; Emi Minamitani et al.	2017	The Journal of Chemical Physics	154	10.1063/1.4997242
Impact of the Local Environment on Li Ion Transport in Inorganic Components of Solid Electrolyte Interphases	Taiping Hu; Jianxin Tian; Fuzhi Dai et al.	2022	Journal of the American Chemical Society	107	10.1021/jacs.2c11521
Thermal conductivity modeling using machine learning potentials: application to crystalline and amorphous silicon	Xin Qian; Shiyu Peng; Xiaobo Li et al.	2019	Materials Today Physics	103	10.1016/j.mtphys.2019.100140
A unified deep neural network potential capable of predicting thermal conductivity of silicon in different phases	Ruiyang Li; Eungkyu Lee; Tengfei Luo	2020	Materials Today Physics	102	10.1016/j.mtphys.2020.100181

Block QC (609 unique)

Title	Authors	Year	Venue	Cited	DOI
Transport and Anderson localization in disordered two-dimensional photonic lattices	Schwartz, Tal; Bartal, Guy; Fishman, Shmuel et al.	2007		1329	10.1038/nature05623
Observation of two-dimensional discrete solitons in optically induced nonlinear photonic lattices	Fleischer, Jason W.; Segev, Mordechai; Efremidis, Nikolaos K. et al.	2003		1039	10.1038/nature01452
Hyperuniform states of matter	S. Torquato	2018	Physics reports	501	10.1016/j.physrep.2018.03.001
Localization and delocalization of light in photonic moiré lattices	Peng Wang; Yuanlin Zheng; Xianfeng Chen et al.	2019	Nature	482	10.1038/s41586-019-1851-6
Decoherence and Disorder in Quantum Walks: From Ballistic Spread to Localization	Schreiber, A.; Cassemiro, K. N.; Potoček, V. et al.	2011		355	10.1103/PhysRevLett.106.180403
Delocalization of a disordered bosonic system by repulsive interactions	B. Deissler; Matteo Zaccanti; G. Roati et al.	2010	Nature Physics	273	10.1038/nphys1635
Topological triple phase transition in non-Hermitian Floquet quasicrystals	Sebastian Weidemann; Mark Kremer; Stefano Longhi et al.	2022	Nature	223	10.1038/s41586-021-04253-0
Observation of topological valley modes in an elastic hexagonal lattice	Vila, Javier; Pal, Raj Kumar; Ruzzene, Massimo	2017		219	10.1103/PhysRevB.96.134307
Disorder-Enhanced Transport in Photonic Quasicrystals	Liad Levi; Mikael C. Rechtsman; Barak Freedman et al.	2011	Science	205	10.1126/science.1202977
Compact localized states and flat-band generators in one dimension	Maimaiti, Wulayimu; Andreanov, Alexei; Park, Hee Chul et al.	2017		184	10.1103/PhysRevB.95.115135

Suggestions

- Block PSC: openalex 4,567, semanticscholar 2,968, ads 2,179, scopus 4,731 hits – a generic term is probably driving this; tighten a group or add a more specific one before reading.
- Block PSC: counts differ >20x across backends (175 to 4,731); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- Block NOV: zero hits on every backend. Either the intersection is genuinely empty (a finding – check the synonyms first) or one group is too narrow; try dropping one group and rerunning.
- Block QC: counts differ >20x across backends (17 to 414); grammar and coverage diverge, so do not compare these numbers – discover here, quote one source.
- 4 block/backend pair(s) hit the `--limit` cap (300); raise it (largest total 4,731) if you need the complete record set rather than the most-cited slice.
- Open-access status was not looked up; rerun with `--pdfs` (optionally `--pdf-blocks`) to collect legal OA PDF links via Unpaywall.
- The PRISMA flow's manual stages are empty: fill in prisma.json in the run directory (screened / excluded / assessed / included) and rerun report.py to complete the diagram.
- Block MLP: total hits changed from 239 (run 20260828T091142) to 838; count drift is expected as indexes grow, but a large jump usually means the query changed – diff queries.json between the runs.